Openi nCol ab

In [ ]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [ ]:
```
df = pd.read_csv('/content/RSS - Kriging.csv')
```

In [ ]:
```
df
```

Out[ ]:

| | X | Y | RSS-A | RSS-B | RSS-C |
|---|---|---|---|---|---|
| 0 | 0 | 2 | 49 | 73 | 69 |
| 1 | 0 | 3 | 50 | 70 | 70 |
| 2 | 0 | 4 | 50 | 68 | 70 |
| 3 | 0 | 5 | 51 | 68 | 71 |
| 4 | 0 | 6 | 52 | 67 | 70 |
| ... | ... | ... | ... | ... | ... |
| 198 | 6 | 26 | 62 | 65 | 51 |
| 199 | 6 | 27 | 61 | 65 | 51 |
| 200 | 6 | 28 | 60 | 65 | 51 |
| 201 | 6 | 29 | 61 | 65 | 51 |
| 202 | 6 | 30 | 61 | 64 | 50 |

203 rows × 5 columns

In [ ]:
```
#Load feature and target
x = df[['X', 'Y']]
y = df [['RSS-A','RSS-B','RSS-C']]
```

In [ ]:
```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(x, y, test_size=0.2, r
```

In [ ]:
```
#Train-Test split
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(x, y, test_size=0.2, r
```

In [ ]:
```
#Train-Model
from sklearn.linear_model import LinearRegression
model=LinearRegression()
model.fit(X_train, y_train)
```

Out[ ]:
```
LinearRegression()
```
In a Jupyter environment, please rerun this cell to show the HTML representation or trust the notebook.

On GitHub, the HTML representation is unable to render, please try loading this page with nbviewer.org.

In [ ]:
```
#prediction
y_pred=model.predict(X_test)
```

In [ ]:
```
#Evaluation
from sklearn.metrics import mean_absolute_error,mean_squared_error,r2_scor
print(mean_absolute_error(y_test, y_pred))
print(mean_squared_error(y_test, y_pred))
print(np.sqrt(mean_squared_error(y_test, y_pred)))
print(r2_score(y_test, y_pred))
```

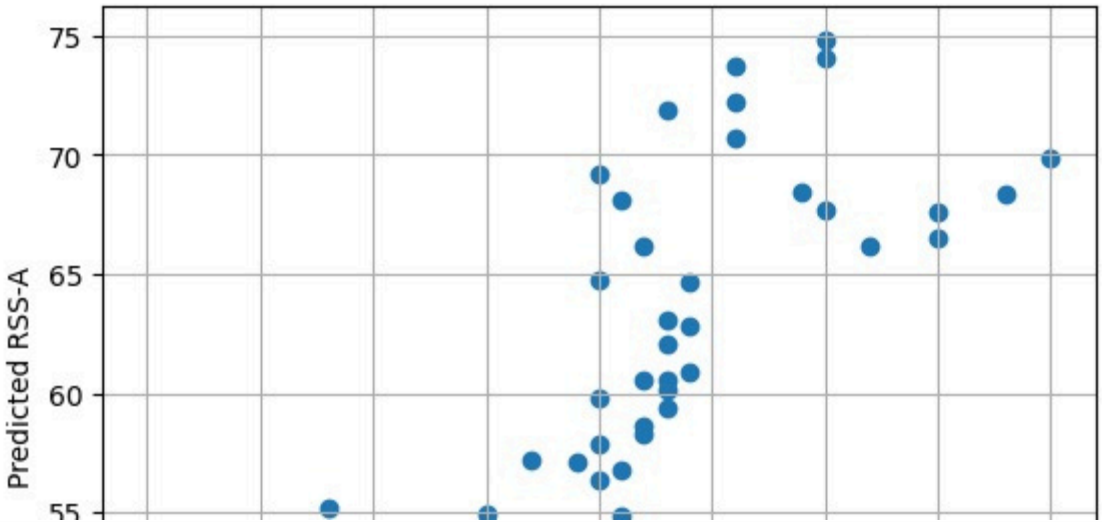
```
2.97339086836549
84
14.81743999709662
3.84934280067346
3
0.6866273567112615
```

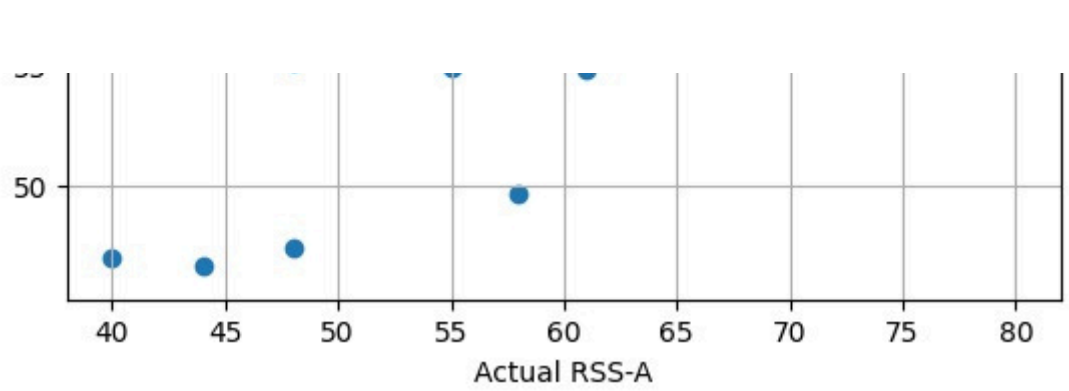
In [ ]:
```
#predict New location
new_location=[[4,3]]
pred=model.predict(new_location)
print(pred)
```

```
[[49.59393161 53.45567818 71.12458586]] /usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but LinearRegression was fitted with feature names
  warnings.warn(
```

In [ ]:
```
#Actual vs Predicted
result=pd.DataFrame({'Actual RSS-A':y_test['RSS-A'].values,'Pred RSS-A':y_
```

In [ ]:
```
#Plot RSS-A
import matplotlib.pyplot as plt
plt.scatter(y_test['RSS-A'],y_pred[:,0])
plt.xlabel('Actual RSS-A')
plt.ylabel('Predicted RSS-A')
plt.grid(True)
plt.show()
```




50
40
45
50
55
60
65
70
75
80
Actual RSS-A